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PICNIC: an open-source Python library for preprocessing of dynamic Positron Emission Tomography (PET) brain imaging data

Mike Schmidt [1,2], **Jeffrey M Miller** [1,2], **Eric Hauser**[1,2], **Dheeraj Puthiya Veedu**[2], **Elizabeth A Bartlett** [1,2], **and Francesca Zanderigo** [1,2]

1 Department of Psychiatry, Columbia University Medical Center, New York, NY, United States **2** Molecular Imaging and Neuropathology Area, New York State Psychiatric Institute, New York, NY, United States

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Summary

PICNIC (Pipeline Initializing Container for Neuro-Imaging Computations) is an open-source Python-based coding library that facilitates ad hoc/non-standardized analyses of brain Positron Emission Tomography (PET) data by performing most standard preprocessing steps for PET data quantification, including brain extraction, tissue segmentation, automatic anatomical parcellation, and motion correction. PICNIC uniquely allows full, transparent modular control over the selection of strategy/software package and associated parameters for each preprocessing step throughout the pipeline. Therefore, the user is provided with the flexibility to select different methods depending on the target of interest and/or radiotracer properties, study population, or other preferences. These settings can be saved as a text-based "input deck" file to support consistent preprocessing across scans, participants, and projects. Furthermore, PICNIC supports freezing software versions within Docker containers, ensuring robust reproducibility.

PICNIC does not require Brain Imaging Data Structure (BIDS) source-level data, but instead takes reconstructed PET static or dynamic data in industry-standard imaging file types (e.g., DICOM, NIfTI, ANALYZE), and converts them to be BIDS-compliant (Knudsen et al., 2020). PICNIC employs the most commonly used brain imaging software packages, including Advanced Normalization Tools (Avants et al., 2008), Analysis of Functional NeuroImages (Cox, 1996), FSL (Jenkinson et al., 2012), FreeSurfer (Fischl, 2012), dcm2niix (Li et al., 2016), and SPM (Functional Imaging Laboratory, 2020). PICNIC was designed with PET in mind, but includes steps involving data from multiple imaging modalities, most prominently structural magnetic resonance imaging (sMRI). PICNIC preprocessing capabilities include image reorientation (to force all images into a standard orientation regardless of scanner defaults), brain extraction (to perform skull-stripping of associated sMRI images), tissue segmentation (to classify sMRI voxels within the brain boundary as grey matter vs. white matter vs. cerebrospinal fluid), automatic anatomical parcellation (to identify regions of interest using any available atlas/parcellation), post-reconstruction rigid-body motion-correction (to correct inter-frame participant motion over the duration of the scan), PET-sMRI co-registration (with 20 different PET-sMRI rigid-body transformations that are automatically ranked based on mutual information), and extraction of time-activity-curves (curves that represent the PET signal over the duration of the scan in delineated regions of interest or in each brain voxel).

PICNIC can be run from the command line or through the provided graphical user interface. An HTML summary output with interactive quality control features allows the user to inspect, comment, and approve/disapprove each module for each participant upon completion. Pre-loaded templates have been provided for commonly used workflows. The user can add, remove

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or edit pre-processing steps to fully customize their pipeline with regards to their dataset. PICNIC currently supports over 100,000 module/parameter combinations.

PICNIC is designed for in vivo brain investigations by expert PET researchers, and has already been applied to preprocess data for recent scientific publications from our Lab (E. Bartlett, Herzog, et al., 2026; E. Bartlett, Zanderigo, et al., 2026; E. Bartlett, Laurell, et al., 2026; E. A. Bartlett, Patil, et al., 2023, 2024; E. A. Bartlett, Zanderigo, Stanley, et al., 2023; E. A. Bartlett, Zanderigo, Ogden, et al., 2023; E. A. Bartlett, Herzog, et al., 2024; E. A. Bartlett, Sublette, et al., 2024; Graves et al., 2024; Herzog et al., 2024, 2025; Laurell, Solingapuram Sai, et al., 2026; Laurell, Bartlett, et al., 2026; Mann et al., 2022; Matheson et al., 2024; Miller et al., 2022; Restifo-Bernstein et al., 2026; F. Zanderigo, Laurell, et al., 2025; F. Zanderigo, Miller, et al., 2025; Francesca Zanderigo et al., 2026) and has been validated externally alongside other pipelines (Haver et al., 2026). PICNIC's modular nature enables extensive customization of preprocessing for a variety of brain PET studies, and its design may allow future extension to preprocessing of data from imaging modalities other than PET. The source code for PICNIC is available at the following link: https://github.com/bil-mind-nyspi/PICNIC.

Statement of Need

PET is a valuable tool used in research and clinical settings to noninvasively image the human brain *in vivo* and can quantify up to nanomolar levels of specific components of brain metabolic and neurochemical processes. Rigorous preprocessing of raw PET data (Nørgaard et al., 2019) is key to obtaining accurate quantification. Although software packages exist that execute each component of brain PET pre-processing, and several of these have even been combined into pipelines, no software exists to support a fully open-source, customizable pre-processing pipeline from start through human quality control (QC).

State of the Field

A variety of software exists for preprocessing brain PET data, but none offer PICNIC's strengths in a single package. PMOD (Mikolajczyk et al., 1998) requires purchase of a license. Both PMOD and CapAIBL (Bourgeat et al., 2015) are closed-source. MIAKAT (Gunn et al., 2016) and MAGIA (Karjalainen et al., 2020) are open-source, but depend on MATLAB, requiring a purchased license. Clinica (Routier et al., 2021) and Neuropypes (Savio et al., 2017) both depend on SPM, which requires a closed-source MATLAB runtime. NiftyPET (Markiewicz et al., 2018) and PETsurfer (Greve et al., 2014, 2016) contain useful PET preprocessing tools, but are not complete pipelines. APPIAN (Funck et al., 2018) is open-source and based on nipype (Gorgolewski et al., 2016), like PICNIC, but uses BEaST brain extraction and ANIMAL segmentation tools, which would create different regions from the commonly used FreeSurfer individualized regions. Tools from PETsurfer are used in PETPrep (Nørgaard et al., 2026) and PETSurfer-bids (Wighton & Greve, 2026), which uses PETPrep output for additional analyses. PETPrep has been developed concurrently with PICNIC, uses nipype to manage steps, and is a reasonable alternative for standard PET preprocessing objectives. PICNIC offers fully open-source and complete pre-processing pipelines, with SPM available but not required, standard FreeSurfer segmentation, and the added ability to customize or extend the pipeline by altering the input deck. Moreover, PICNIC writes an HTML-based report with interactive image viewers, images, and videos to guide QC.

Software Design

PICNIC was written with Python3 and uses nipype to orchestrate the execution of well-supported external neuroimaging utilities like ANTs (Avants et al., 2008), AFNI (Cox, 1996), Connectome Workbench (Marcus et al., 2011), FreeSurfer, PETPVC (Thomas et al., 2016),

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and FSL. PICNIC is intended to be run as a Docker container to provide all dependencies pre-built in a working environment. However, PICNIC and its dependencies can alternatively be installed and run locally. PICNIC also implements the interactive Papaya neuroimaging viewer (Institute, 2019) alongside prepared images and video within its HTML reports to guide QC, supporting thorough manual review.

AI Usage Disclosure

AI was used to find and correct errors in autodoc scripting for creation of documentation. AI was used to verify bibtex citations in this document. AI has not been used to create or maintain PICNIC code or in the writing of this document.

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